

HUJI Applicative Research Report — Exact & Social Sciences — Week 40, 2026

2 paper(s) · Exact & Social Sciences · Weekly · Week 40, 2026 · Generated September 28, 2026

■ ■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

This week's digest highlights research poised to tackle critical challenges in clean energy and public health. One breakthrough delves into the fundamental mechanisms of next-generation sodium-ion batteries, promising more sustainable and cost-effective grid-scale energy storage solutions. Concurrently, another study offers vital insights into the unique mental health needs and care barriers faced by diverse adolescent populations, paving the way for highly personalized digital health interventions. These distinct yet impactful investigations underscore Hebrew University's commitment to innovation with significant societal and economic relevance.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

- Daphna Shimon — Materials, Clean Energy
- Yehuda Neumark — Software/AI, Clinical, Other

EXACT & SOCIAL SCIENCES — RESEARCH HIGHLIGHTS

1. Elucidating Sodium-Ion Storage Mechanisms in Bowl-Shaped Hierarchical Porous Carbon Anodes: A Combined Solid-State NMR and EPR Approach

23/50

"Groundbreaking insights into sodium-ion battery materials promise cheaper, more sustainable grid-scale energy storage."

Main researcher: Daphna Shimon · daphna.shimon@mail.huji.ac.il

Institute of Chemistry , The Hebrew University of Jerusalem , ,

Published · 2026 Sep 26

ABOUT THIS RESEARCH

This research uses advanced spectroscopic techniques to understand how sodium ions move and store energy within specialized porous carbon structures. By mapping these microscopic mechanisms, researchers can better design high-performance electrodes for next-generation battery technologies.

COMMERCIAL ANGLE

The ability to engineer hierarchical porous carbons provides a pathway to developing cheaper, more sustainable sodium-ion batteries that can compete with lithium-ion for grid-scale energy storage.

COMMERCIALISATION METRICS

Novelty

4/10

The research appears to be an incremental mechanistic study using standard spectroscopic techniques (NMR/EPR) to characterize a known material class (porous carbon) for sodium-ion batteries. While scientifically sound, it lacks the platform-enabling or paradigm-shifting characteristics required for a high novelty score in deep-tech investing.

Commercial Potential

4/10

The paper focuses on fundamental mechanistic characterization using high-end spectroscopy rather than demonstrating a scalable manufacturing process or a specific battery cell performance metric. While the insights could optimize future materials, the work is foundational science rather than a direct product or platform technology.

Market Size

9/10

Sodium-ion battery technology targets the massive global energy storage and grid-scale battery markets, offering a lower-cost alternative to lithium-ion for stationary storage and mass-market mobility.

Tech Readiness

3/10

The work is focused on fundamental mechanistic elucidation using advanced spectroscopic techniques (NMR and EPR), indicating basic research rather than prototype development or scaling. It resides at the fundamental science stage (TRL 2-3) intended to inform future material design.

IP Strength

3/10

The paper appears to be fundamental characterization research focused on elucidating scientific mechanisms rather than disclosing a novel, scalable material composition or a specific manufacturing process. While the insights could inform future IP, the work itself is descriptive and mechanistic, offering low immediate defensibility.

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2. Latent profiles of mental health and care barriers among heterosexual and sexual minority adolescents.12/50

"Understanding adolescent mental health profiles enables personalized digital solutions for underserved young populations."

Main researcher: Yehuda Neumark · yehuda.neumark@mail.huji.ac.il
Braun School of Public Health and Community Medicine, Hebrew University of Jerusalem, Jerusalem, Israel.
Published in Israel journal of health policy research · 2026-09

ABOUT THIS RESEARCH

This research examines the psychological profiles and specific obstacles to accessing healthcare among heterosexual and sexual minority adolescents. It aims to identify distinct patterns of mental health needs and systemic barriers faced by these diverse demographic groups.

COMMERCIAL ANGLE

There is a commercial opportunity for digital health platforms and personalized mental health software that utilize psychometric profiling to tailor interventions for marginalized adolescent populations.

COMMERCIALISATION METRICS

Novelty3/10

The study appears to be an observational epidemiological or sociological analysis rather than a technical or methodological innovation. It likely applies existing statistical frameworks to categorize known risk factors, offering incremental clinical insight rather than a groundbreaking scientific or technological leap.

Commercial Potential2/10

This is a descriptive epidemiological or sociological study focused on population stratification rather than the development of a proprietary tool, therapeutic, or diagnostic technology. While the findings may inform public health policy, they lack a direct path to a protectable product or scalable commercial service.

Market Size4/10

The paper focuses on descriptive epidemiological profiling rather than a therapeutic intervention or technology, making it a research study rather than a scalable product or platform. While the subject matter (adolescent mental health) addresses a massive societal need, the output itself does not define a clear commercial market or addressable unit of demand.

Tech Readiness2/10

This is a descriptive observational study identifying sociological and psychological patterns rather than the development of a technical tool or intervention. It represents foundational basic research that informs potential future applications but lacks a functional technological prototype.

IP Strength1/10

The paper describes an observational social science study on population demographics and healthcare barriers, which lacks any patentable technology, proprietary algorithms, or hardware inventions.

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